Unifying Flight Control and Mission Computing: A Single-Chip Architecture for Autonomous Drones

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Abstract

The market for high-performance Unmanned Aerial Vehicles (UAVs) is dominated by proprietary, closed-ecosystem solutions that rely on custom ASICs to integrate flight control and mission computing. This creates a significant barrier to entry for the open-source community and researchers, who are often constrained by the performance bottlenecks and SWaP-C (Size, Weight, Power, and Cost) penalties of conventional two-board (flight controller and companion computer) architectures. This paper introduces a novel, open-source avionics platform that architecturally converges these disparate functions onto a single, power-efficient FPGA-based System-on-Chip (SoC). We propose a solution built on the Microchip PolarFire SoC, which features a asymmetric RISC-V CPU cluster and a flexible FPGA fabric. This enables a mixed-criticality system running both the NuttX real-time operating system for deterministic flight control and a full Linux environment for high-level computation on the same die. This unified system serves as a powerful development platform for rapidly prototyping and deploying the computationally intensive algorithms required for autonomous flight.

1 Introduction

Unmanned Aerial Vehicles (UAVs) are increasingly used in applications ranging from agriculture and public safety to autonomous logistics. DJI dominates the consumer drone market with over 70% share [1], but its proprietary ASIC-based platforms restrict flexibility, limiting research and innovation.

Many UAV research projects use Single Board Computers (SBCs) with flight controllers, but SBCs run non-deterministic OSes, adding SWaP overhead and making them unsuitable for hard real-time tasks. Architectural separation further complicates integration of safety-critical control and high-performance computing, limiting their usefulness in advanced UAV research [2, 3].

Recent works have demonstrated the potential of SoC-FPGA platforms in UAV avionics—for example, accelerating autonomous landing vision pipelines on Zynq devices [4], hybrid FPGA–DSP flight controllers [5], and Aerotenna's SoC-based flight controller for ArduPilot [6]. However, these remain fragmented or closed solutions, without integration of both the *flight computer* (Linux-class) and *flight controller* (RTOS-class) into a single, open platform.

This work proposes an **open-source, integrated avionics SoC** that addresses size, weight, power, and deterministic real-time performance. Using the **Microchip PolarFire SoC FPGA**, NuttX provides hard real-time control, Linux handles autonomy, and the FPGA fabric accelerates sensor fusion and perception tasks. This unified approach reduces system overhead and enables a reconfigurable, community-driven ecosystem for next-generation UAVs.

2 Main Idea

The project proposes the integration of the flight controller and the flight computer typically used in conventional drones. There is a move away from systems built from discrete components and toward highly integrated solutions like SoCs and Systems-in-Package (SiPs). This trend of miniaturization and functional integration is particularly pronounced in the aerospace and defense sectors. The core motivations for this approach are:

1. Optimization of Size, Weight, Power, and Cost (SWaP-C).
2. Creation of a testbed for architectural-level implementations of algorithms required for autonomous flight.
3. Development of a more secure communication system.
4. Create an Open Source Ecosystem based on PX4, Ubuntu and ROS for creating an aerial robotics testbed

2.1 Platform Selection

The Microchip Polarfire SoC Icicle Kit consists of 5 cores (1 Monitor Core and 4 General Purpose Cores). We propose on using this SoC due to its support for Asymmetric Multi-Processing. This allows us to boot each of the cores with a different OS, allowing the flight control firmware and the flight computer OS to run simultaneously.

2.2 Firmware Selection

PX4 was selected for the PolarFire SoC due to its reliance on the NuttX RTOS, which has native support on the platform. This avoided the complex OS porting required by alternatives.

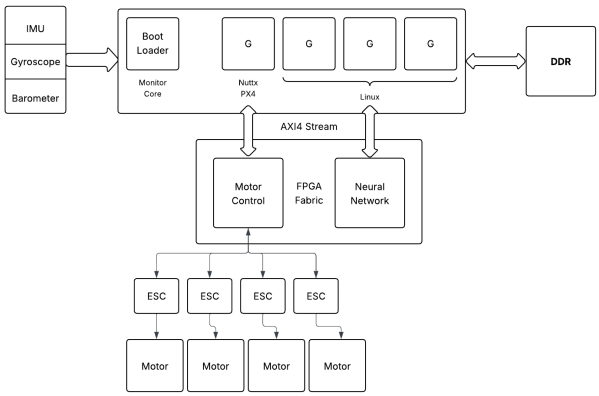


Figure 1: Block Diagram of the system

2.3 Core Configuration

The monitor E51 core runs the bootloader. One U54 Core will be dedicated to running the Apache NuttX RTOS. The remaining three U54 application cores will run a full Linux distribution in a symmetric multiprocessing (SMP) configuration.

2.4 Inter-core Communication

NuttX and Linux communicate through the OpenAMP framework, which enables standardized inter-processor communication via shared memory. This communication is facilitated by the RPMsg component, which provides a reliable channel between the cores.

3 Application

The proposed system offers a comprehensive solution for accelerating the entire lifecycle of various algorithms for an autonomous UAV. This offers the user a full-fledged flight controller thereby creating a tested for the computationally intensive algorithms which are required for autonomous flight.

This system can mainly be used to explore various architectures. These can be used to check which algorithms can run better on FPGA and on CPU cores. We can obtain results for power, performance, and area optimization.

4 Value Addition

In the conventional two-board architecture, the real-time flight controller and the high-level companion computer are connected by a slow serial connection. The bottleneck is overcome with high-bandwidth, low-latency on-chip memory bus. This reduces communication latency by orders of magnitude.

The conventional model presents a large security risk, as the network-connected companion computer is a common entry point for attackers. This design shrinks the physical attack surface.

The primary advantage of this system is that the 5 core CPU and the FPGA are integrated within the SoC which enables intensive tasks to be offloaded to the FPGA and the less intensive tasks to be done by the CPU which reduces the overall processing time and enables real time execution of complex flight control algorithms.

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